

C - Understory

分值: 300 分

Problem Statement

Takahashi is managing the number of trees in his garden. Initially, there are no trees in the garden.

高桥正在管理他花园里的树木数量。初始时，花园里没有树。

Q queries are given in order. Each query is one of the following two types. Immediately after processing each query, output the number of trees currently in the garden.

按顺序给出 Q 次查询，每次查询属于以下两种类型之一。处理完每次查询后，立即输出当前花园中的树木数量。

- 1 h : Add one new tree of height h to the garden.
- 2 h : Remove all trees currently in the garden whose height is at most h .
- 1 h : 向花园中添加一棵高度为 h 的新树。
- 2 h : 移除花园中当前所有高度不超过 h 的树。

Constraints

- $1 \leq Q \leq 3 \times 10^5$
- $1 \leq h \leq 10^9$
- All input values are integers.
- $1 \leq Q \leq 3 \times 10^5$
- $1 \leq h \leq 10^9$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

Q
 query_1
 query_2
 $\vdots$
 query_Q

query_i , representing the i -th query, is one of the following two types:

query_i 代表第 i 次查询，属于以下两种类型之一：

$1\ h$
“
“*text*
 $2\ h$

Output

Output Q lines.

输出 Q 行。

The i -th line should contain the number of trees in the garden immediately after processing the i -th query.

第 i 行应包含处理完第 i 次查询后，花园中当前的树木数量。

Sample Input 1

```
5
1 5
1 7
1 8
2 7
1 3
```

Sample Output 1

```
1
2
3
1
2
```

The number of trees changes as follows.

树木数量的变化过程如下：

- A tree of height 5 is added. The garden contains one tree of height 5.
- A tree of height 7 is added. The garden contains two trees of heights 5, 7.
- A tree of height 8 is added. The garden contains three trees of heights 5, 7, 8.
- Trees of height at most 7 are removed. The garden contains one tree of height 8.
- A tree of height 3 is added. The garden contains two trees of heights 8, 3.
- 添加一棵高度为 5 的树，花园中有1棵高度为 5 的树。
- 添加一棵高度为 7 的树，花园中有2棵高度分别为 5, 7 的树。
- 添加一棵高度为 8 的树，花园中有3棵高度分别为 5, 7, 8 的树。
- 移除所有高度不超过 7 的树，花园中有1棵高度为 8 的树。
- 添加一棵高度为 3 的树，花园中有2棵高度分别为 8, 3 的树。

Sample Input 2

```
12
2 256601193
1 85138616
1 202564041
2 276477192
1 55551662
1 170271057
2 754166580
1 854388209
1 772036624
2 651124113
```

```
1 301137866
2 290875185
```

Sample Output 2

```
0
1
2
0
1
2
0
1
2
2
3
3
```